Telemetry Validation

AIO · MDM · VALIDATION REPORT

20260925-164942-without-037

Firmware-client vitals as the MDM stored them, graded against adb on the same devices.

4h test · without **AT070AA2600037** (v2.1.017, client 1.0.0) · **4** device(s) · **16:49:44–20:49:45** (240 min) · clock offset **+0.1s**
mdm.dev.aioapp.com + mdm-stage.dev.aioapp.com

At a glance · all devices

11 of 12 fields match on every device. 1 needs attention.

FAIL Battery temperature — within ±1 °C only 96% of the time live, 90% in chart history (typically 0.2 °C off, worst 1.8 °C), on 2 of 4 devices

DEVICES	ACCURATE READINGS	DASHBOARD BEHIND DEVICE	TEST	READ ERRORS
4 4 on live	**99.8%** 1153 of 1155 checks, all fields	**3.6 min** 95% of the time under 8.8 min	**240 min** sampled every 60 s	**1** 0 adb of 964 · 1 API of 963

FIELD	ALLOWED	RESULT	LIVE VALUE	CHART HISTORY	TYPICAL ERROR	WORST ERROR	DEVICES OFF
Battery level `battery_pct`	±0 %	PASS	100.0% (51/51)	100.0% (50/50)	±0%	±0%	none
Charging state `charging`	exact	PASS	100.0% (51/51)	100.0% (50/50)	—	—	none
Charger type `charger_type`	exact	PASS	100.0% (51/51)	—	—	—	none
Battery present `battery_present`	exact	PASS	100.0% (51/51)	—	—	—	none
Battery voltage `charger_voltage_mv`	±75 mV	PASS	100.0% (51/51)	—	±10 mV	±35 mV	none
Battery temperature `battery_temp_c`	±1 °C	FAIL	96.1% (49/51)	90.0% (45/50)	±0.2 °C	±1.8 °C	2 of 4 AT070AABU00231, AT070AABU00280
Wireless charging `wlc_charging`	exact	PASS	100.0% (51/51)	—	—	—	none
Wireless pad status `wlc_status`	exact (1 s watcher, 120 s cache)	PASS	100.0% (165/165)	100.0% (162/162)	—	—	none
RAM total `ram_total_mb`	±1 MB	PASS	100.0% (156/156)	100.0% (152/152)	±0 MB	±0 MB	none
RAM available `ram_available_mb`	±5 % of total	PASS	100.0% (156/156)	—	±1 MB	±30 MB	none
RAM used `ram_used_mb`	±5 % of total	PASS	100.0% (156/156)	100.0% (152/152)	±2 MB	±31 MB	none
Free storage `storage_free_gb`	±0.1 GB	PASS	100.0% (165/165)	100.0% (162/162)	±0 GB	±0.1 GB	none

Live value: the MDM's current value checked against adb when the device built it. Chart history: the stored rows behind the dashboard charts. Pass = 100%, Watch = 95% or more, Fail = under 95%. Errors are measured against the closest adb reading in the field's cache window.

Tolerances

Change an allowance and the summary above and sections 1 and 3 below re-grade on the spot. The miss lists below each table stay at the default rules. The link keeps your values, so it can be shared.

battery_pct	charger_voltage_mv	battery_temp_c	ram_total_mb	ram_available_mb	ram_used_mb	storage_free_gb	
± 0 %	± 75 mV	± 1.0 °C	± 1 MB	± 5 % of total	± 5 % of total	± 0.1 GB	Reset to defaults

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Details

Per device and per field. The summary above is built from these.

Devices

SERIAL	MDM	ADB TARGET	ADB SAMPLES	ADB ERRORS	API READS	API ERRORS	DISTINCT FRAMES	FRAME AGE P50 / P95 / MAX (CHARGING)	(ON BATTERY)
AT070AABU00231	live	10.32.3.29:5555	241	0	241	0	33	—	4.3m / 9.0m / 10.2m
AT070AABU00280	live	10.32.0.251:5555	241	0	241	0	34	—	4.0m / 9.0m / 10.2m
AT070AABU00833	live	10.32.3.30:5555	241	0	241	0	37	—	4.0m / 9.0m / 10.4m
AT070AABU00875	live	10.32.3.31:5555	241	0	240	1	65	—	2.3m / 7.2m / 9.8m

1 · Accuracy at frame time

Each distinct frame, placed on the adb timeline by its `uptime_seconds`, checked against the adb snapshots within the field's cache window. Anything under 100% here is a real mismatch. Under each rate: the observed deviation (MDM - adb, against the best-matching snapshot in the window): its range, its median size and its largest size, to compare with the Allowed column.

FIELD	ALLOWED	AT070AABU00231	AT070AABU00280	AT070AABU00833	AT070AABU00875
battery_pct	±0 %	100.0% (7/7) observed ±0 % (all exact)	100.0% (12/12) observed ±0 % (all exact)	100.0% (19/19) observed ±0 % (all exact)	100.0% (13/13) observed ±0 % (all exact)
charging	exact	100.0% (7/7)	100.0% (12/12)	100.0% (19/19)	100.0% (13/13)
charger_type	exact	100.0% (7/7)	100.0% (12/12)	100.0% (19/19)	100.0% (13/13)
battery_present	exact	100.0% (7/7)	100.0% (12/12)	100.0% (19/19)	100.0% (13/13)
charger_voltage_mv	±75 mV	100.0% (7/7) observed -11…+18 mV · median ±11 mV · max ±18 mV	100.0% (12/12) observed +19…+35 mV · median ±29.5 mV · max ±35 mV	100.0% (19/19) observed -15…+6 mV · median ±6 mV · max ±15 mV	100.0% (13/13) observed -19…+3 mV · median ±8 mV · max ±19 mV
battery_temp_c	±1 °C	85.7% (6/7) observed -0.4…+1.6 °C · median ±0.3 °C · max ±1.6 °C	91.7% (11/12) observed -0.3…+1.8 °C · median ±0.15 °C · max ±1.8 °C	100.0% (19/19) observed -0.6…+0.6 °C · median ±0.2 °C · max ±0.6 °C	100.0% (13/13) observed -0.2…+1 °C · median ±0 °C · max ±1 °C
wlc_charging	exact	100.0% (7/7)	100.0% (12/12)	100.0% (19/19)	100.0% (13/13)
wlc_status	exact (1 s watcher, 120 s cache)	100.0% (32/32)	100.0% (33/33)	100.0% (36/36)	100.0% (64/64)
ram_total_mb	±1 MB	100.0% (27/27) observed ±0 MB (all exact)	100.0% (32/32) observed ±0 MB (all exact)	100.0% (34/34) observed ±0 MB (all exact)	100.0% (63/63) observed ±0 MB (all exact)
ram_available_mb	±5 % of total	100.0% (27/27) observed -23…+3 MB · median ±1 MB · max ±23 MB (0.6% of total)	100.0% (32/32) observed -7…+20 MB · median ±1.5 MB · max ±20 MB (0.6% of total)	100.0% (34/34) observed -15…+22 MB · median ±1 MB · max ±22 MB (0.6% of total)	100.0% (63/63) observed -12…+30 MB · median ±1 MB · max ±30 MB (0.8% of total)
ram_used_mb	±5 % of total	100.0% (27/27) observed -4…+22 MB · median ±1 MB · max ±22 MB (0.6% of total)	100.0% (32/32) observed -21…+7 MB · median ±2 MB · max ±21 MB (0.6% of total)	100.0% (34/34) observed -23…+14 MB · median ±1 MB · max ±23 MB (0.6% of total)	100.0% (63/63) observed -31…+12 MB · median ±2 MB · max ±31 MB (0.9% of total)
storage_free_gb	±0.1 GB	100.0% (32/32) observed ±0 GB (all exact)	100.0% (33/33) observed ±0 GB (all exact)	100.0% (36/36) observed ±0 GB (all exact)	100.0% (64/64) observed -0.1…+0.1 GB · median ±0 GB · max ±0.1 GB

AT070AABU00231 misses
(first few per field)

TIME	FIELD	MDM	ADB (NEAREST)
17:02:43	battery_temp_c	26.6°C	25.0°C

AT070AABU00280 misses
(first few per field)

TIME	FIELD	MDM	ADB (NEAREST)
17:12:43	battery_temp_c	26.4°C	24.6°C

2 · Freshness: how long the dashboard disagreed with the device

% of observed time the server's current value was outside tolerance of the device's current value, and the longest such stretch. Expected on battery: the client stretches its poll to ≥5 min while the WebSocket is healthy.

FIELD	AT070AABU00231	AT070AABU00280	AT070AABU00833	AT070AABU00875
battery_pct	0.0%	0.8% · longest 0s at 19:10:48 (battery: MDM 44% vs 43%)	2.1% · longest 0s at 16:49:51 (battery: MDM 35% vs 34%)	0.4% · longest 0s at 16:54:48 (battery: MDM 55% vs 54%)
charging	0.0%	0.0%	0.0%	0.0%
charger_type	0.0%	0.0%	0.0%	0.0%
battery_present	0.0%	0.0%	0.0%	0.0%
charger_voltage_mv	0.0%	0.0%	0.0%	0.0%
battery_temp_c	1.7% · longest 2.0m at 17:00:46 (battery: MDM 26.3°C vs 25.1°C)	0.8% · longest 0s at 17:12:46 (battery: MDM 26.4°C vs 24.6°C)	0.0%	2.9% · longest 118s at 16:54:48 (battery: MDM 23.2°C vs 24.7°C)
wlc_charging	0.0%	0.0%	0.0%	0.0%
wlc_status	0.0%	0.0%	0.0%	0.0%
ram_total_mb	0.0%	0.0%	0.0%	0.0%
ram_available_mb	2.5% · longest 5.0m at 16:54:47 (battery: MDM 1685 MB vs 1896 MB)	2.9% · longest 6.0m at 16:55:47 (battery: MDM 1706 MB vs 1900 MB)	2.1% · longest 4.0m at 16:55:46 (battery: MDM 1661 MB vs 1873 MB)	0.0%
ram_used_mb	2.5% · longest 5.0m at 16:54:47 (battery: MDM 1944 MB vs 1734 MB)	2.9% · longest 6.0m at 16:55:47 (battery: MDM 1923 MB vs 1730 MB)	2.1% · longest 4.0m at 16:55:46 (battery: MDM 1969 MB vs 1757 MB)	0.0%
storage_free_gb	0.0%	0.0%	0.0%	0.0%

3 · History rows (what the charts and CSV read)

The stored samples returned by `GET /api/v1/devices/{serial}` inside the collection window, graded the same way. Row gaps below.

FIELD	ALLOWED	AT070AABU00231	AT070AABU00280	AT070AABU00833	AT070AABU00875
battery_pct	±0 %	100.0% (10/10) observed ±0 % (all exact)	100.0% (13/13) observed ±0 % (all exact)	100.0% (18/18) observed ±0 % (all exact)	100.0% (9/9) observed ±0 % (all exact)
charging	exact	100.0% (10/10)	100.0% (13/13)	100.0% (18/18)	100.0% (9/9)
charger_type	exact	—	—	—	—
battery_present	exact	—	—	—	—
charger_voltage_mv	±75 mV	—	—	—	—
battery_temp_c	±1 °C	80.0% (8/10) observed -0.4…+1.6 °C · median ±0.35 °C · max ±1.6 °C	76.9% (10/13) observed -0.2…+1.8 °C · median ±0.1 °C · max ±1.8 °C	100.0% (18/18) observed -0.6…+0.6 °C · median ±0.2 °C · max ±0.6 °C	100.0% (9/9) observed -0.2…+1 °C · median ±0 °C · max ±1 °C
wlc_charging	exact	—	—	—	—
wlc_status	exact (1 s watcher, 120 s cache)	100.0% (31/31)	100.0% (33/33)	100.0% (35/35)	100.0% (63/63)
ram_total_mb	±1 MB	100.0% (26/26) observed ±0 MB (all exact)	100.0% (31/31) observed ±0 MB (all exact)	100.0% (34/34) observed ±0 MB (all exact)	100.0% (61/61) observed ±0 MB (all exact)
ram_available_mb	±5 % of total	—	—	—	—
ram_used_mb	±5 % of total	100.0% (26/26) observed -4…+22 MB · median ±1 MB · max ±22 MB (0.6% of total)	100.0% (31/31) observed -21…+7 MB · median ±2 MB · max ±21 MB (0.6% of total)	100.0% (34/34) observed -23…+14 MB · median ±1.5 MB · max ±23 MB (0.6% of total)	100.0% (61/61) observed -31…+12 MB · median ±1 MB · max ±31 MB (0.9% of total)
storage_free_gb	±0.1 GB	100.0% (31/31) observed ±0 GB (all exact)	100.0% (33/33) observed ±0 GB (all exact)	100.0% (35/35) observed ±0 GB (all exact)	100.0% (63/63) observed -0.1…+0.1 GB · median ±0 GB · max ±0.1 GB

AT070AABU00231 misses
(first few per field)

TIME	FIELD	MDM	ADB (NEAREST)
17:02:46	battery_temp_c	26.6°C	25.0°C
19:54:46	battery_temp_c	24.9°C	23.4°C

AT070AABU00280 misses
(first few per field)

TIME	FIELD	MDM	ADB (NEAREST)
17:12:47	battery_temp_c	26.4°C	24.6°C
17:12:45	battery_temp_c	26.4°C	24.6°C
18:27:46	battery_temp_c	26.0°C	24.5°C

SERIAL	ROWS	GAP P50	GAP P95	GAP MAX
AT070AABU00231	31	9.0m	10.3m	10.5m
AT070AABU00280	33	8.9m	10.2m	10.3m
AT070AABU00833	35	8.6m	10.3m	10.5m
AT070AABU00875	63	3.0m	8.3m	10.4m

Known client behaviour (not bugs in this test)

- `battery_pct` is the level the firmware broadcasts: `BatteryService` remaps 90–98% to 90–100% for display, so `dumpsys battery` (raw) can read 99 while the status bar and the MDM say 100. adb values are remapped the same way before comparing.
- `charger_voltage_mv` is `EXTRA_VOLTAGE` — the battery voltage, not the charger's.
- `battery_temp_c` is healthd's reading as-is (client 1.4.8+), graded against adb's; clients 1.4.6–1.4.7 reported a 3-minute average and will miss on every jump. RAM is cached 30 s, storage 120 s.
- On battery with a healthy WebSocket the client polls every ≥5 min; charging edges and WLC changes are pushed immediately.
- The server stores no device timestamp; this report times frames by `uptime_seconds` vs `/proc/uptime`.